

Sycophancy Under Deliberative Scaffolding: Construct Fragmentation, Judge Gaming, and Panel-Robust Prompt Optimization

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Abstract

“Sycophancy” names several behaviours, and a mitigation that targets one need not touch the others. We study a black-box deliberative scaffold—Narration-of-Thought (NoT), a system prompt that forces a model to enumerate stakeholders, project consequences, state uncertainty, and commit before answering (specified in §3)—and ask which form of sycophancy it moves. **(1) Construct fragmentation.** On social-sycophancy advice tasks (Cheng et al., 2025), NoT cuts emotional-validation rates on all four frontier generators (Fisher $p < 10^{-4}$; -26 to -53 pp vs. chain-of-thought), landing within two points of the human rate on three of four; but it does *not* transfer to propositional sycophancy on false-theorem proving (Petrov et al., 2025) (significant reduction on one generator; a significant *increase* on another) and *reverses* on cross-prompt moral consistency. Hand-written obvious-falsehood probes (Sharma et al., 2024) are saturated ($\leq 2.2\%$), making BrokenMath (39–83%) the appropriate propositional instrument. **(2) Single-judge prompt optimization games the judge.** Optimizing the scaffold against one absolute LLM judge with textual gradients drives held-out validation to 0.7% and beats matched-budget TextGrad, OPRO and APE—but a three-family panel flags all three social metrics as low inter-judge agreement (Krippendorff’s $\alpha \leq 0.42$), and re-scoring the *identical* responses exposes the indirectness reduction as a Goodhart artifact (6% under the training judge, 87% under a held-out judge). **(3) A panel-robust optimizer.** Replacing the single-judge loss with a worst-case-over-panel objective, one judge family held out entirely, yields a prompt that generalizes to the held-out family significantly better than single-judge optimization (framing 11% vs. 52%, $p=1.9 \times 10^{-14}$; indirectness 4% vs. 74% on out-of-family DeepSeek-R1, $p < 10^{-15}$) while preserving stakeholder depth. We conclude that deliberative scaffolding tar-

gets *social*, *single-response* face-preservation, and that prompt optimization against LLM judges must be evaluated against judges held out from optimization—or it measures judge idiosyncrasy rather than behaviour.

1 Introduction

Sycophancy—an assistant agreeing with or flattering a user at the expense of accuracy or candour—is a recognised consequence of preference-based fine-tuning (Sharma et al., 2024; Perez et al., 2023), visible in the GPT-4o rollback of 2025 (OpenAI, 2025) and a growing benchmark literature (Malmqvist, 2024). Mitigation work often treats sycophancy as a single axis, yet the construct is now understood to fragment. *Propositional* sycophancy is endorsing checkably false claims; *social* sycophancy is excessive preservation of the user’s “face”—their desired self-image—through validation, indirect language, accepting framing, or moral affirmation (Cheng et al., 2025; Goffman, 1955). These are measured by different instruments and need not respond to the same intervention; indeed Cheng et al. (2025) already observe that the linguistic facets of social sycophancy are more mitigable than the reasoning-laden ones.

We study whether a single inference-time scaffold closes any of these gaps. *Narration-of-Thought* (NoT) is a system prompt—requiring no fine-tuning, weights, or activations—that, in place of “think step by step,” asks the model to name a protagonist, enumerate the stakeholders whose interests intersect the decision, project consequences two steps out, articulate what remains uncertain, and only then commit to an answer (Anonymous, 2025). We specify NoT in full in §3; the reader need not consult prior work. NoT is, by construction, a black-box, inference-time, output-side intervention, which distinguishes it from white-box steering (Cheng et al., 2025), training-side finetun-

ing (Wei et al., 2023; Khan et al., 2024), and the recent black-box *input-side* reframing of Dubois et al. (2026).

Two facts make this a measurement problem as much as a mitigation problem. Every sycophancy outcome here is scored by an LLM judge, and LLM judges carry self-preference, verbosity, and instability biases with chronically low inter-judge agreement (Zheng et al., 2023; Panickssery et al., 2024; Li et al., 2024); and once a prompt is *optimized* against such a judge, the setup is a textbook case of reward over-optimization—a proxy that becomes a target ceases to measure what it did (Strathern, 1997; Gao et al., 2023; Pan et al., 2024). We confront both directly.

Contributions. (1) We document obvious-falsehood saturation and adopt BrokenMath as the rigorous propositional instrument (§4). (2) We show NoT reduces social validation and framing across four generators but dissociates from propositional error-detection and from cross-prompt moral consistency—direct evidence that sycophancy is not one construct (§5–§7). (3) We separate two moral measurements (binary-suffix vs. free-form verdict extraction) and retract an earlier “moral-collapse” reading traceable to a data-loading error (§6). (4) We show single-judge prompt optimization Goodharts its judge—a reduction that vanishes on held-out judges—using a cross-family reliability panel and a same-responses re-scoring diagnostic (§8). (5) As our methods contribution, we introduce a black-box *panel-robust* optimizer minimizing a worst-case-over-judges loss that generalizes to a held-out, out-of-family judge significantly better than single-judge optimization (§9).

2 Related Work

Sycophancy and its constructs. Perez et al. (2023) and Sharma et al. (2024) establish propositional sycophancy and trace it to preference data; SycophancyEval is the standard obvious-falsehood instrument. Plausible-falsehood benchmarks find the residual sycophancy it misses—SycEval reports 58% capitulation (Fanous et al., 2025) and BrokenMath finds the strongest model sycophantic 29% of the time on expert-verified false theorems (Petrov et al., 2025). Cheng et al. (2025) introduce *social* sycophancy and the ELEPHANT benchmark (validation, indirectness, framing, moral), grounded in face-work (Goff-

man, 1955); multi-turn (Hong et al., 2025) and advice-seeking (Kim et al., 2025) settings extend it. Our result sharpens an asymmetry ELEPHANT already noted into a dissociation under one intervention.

Sycophancy mitigation. Training-side methods include synthetic-data finetuning (Wei et al., 2023) and preference optimization (Khan et al., 2024); white-box methods include steering (Cheng et al., 2025) and probe penalties (Papadatos and Freedman, 2024). Closest to NoT are black-box inference-time methods: input-side question-reframing, which Dubois et al. (2026) show beats the naive “don’t be sycophantic” prompt, and trace monitoring of reasoning-model sycophancy (Hu et al., 2025). NoT occupies the remaining cell— inference-time, black-box, *output-side* deliberation. A recurring caveat is overcorrection: such interventions can make models reject *correct* user claims (Wei et al., 2023; Sharma et al., 2024), which we revisit in §10.

Chain-of-thought and judges. NoT is a constrained chain-of-thought (Wei et al., 2022; Kojima et al., 2022); added reasoning need not reduce bias—CoT rationalizes user-cued answers without verbalizing them (Turpin et al., 2023), requiring consistency training to fix (Chua et al., 2024), which predicts deliberation helping some constructs and backfiring on others. Since every outcome is judge-scored, judge reliability is load-bearing: LLM judges reach only ~63–66% human agreement (Zheng et al., 2023; Dubois et al., 2023), show self-preference (Panickssery et al., 2024), verbosity/position biases (Li et al., 2024; Gu et al., 2024), and instability under prompt perturbation (Thakur et al., 2024). We use Krippendorff’s α (Krippendorff, 2004) and Cohen’s κ (Cohen, 1960); our “high κ -vs-human, low inter-judge α ” pattern is the base-rate κ paradox (Feinstein and Cicchetti, 1990), and our panel adopts established majority-vote aggregation.

Prompt optimization and reward hacking. Prompt optimizers include APE (Zhou et al., 2023), OPRO (Yang et al., 2024), and the textual-gradient methods ProTeGi (Pryzant et al., 2023) and TextGrad (Yuksekgonul et al., 2024) (cf. DSPy (Khatab et al., 2024)); our “narrative gradients” are a construct-specialized textual-gradient optimizer. Optimizing a learned proxy eventually diverges from truth (Strathern, 1997; Amodei

et al., 2016; Skalse et al., 2022; Gao et al., 2023), and *in-context* reward hacking arises when a model maximizes evaluation scores without weight updates (Pan et al., 2024)—our single-judge regime. Reward-model *ensembles* mitigate over-optimization by averaging (Coste et al., 2024; Eisenstein et al., 2023; Ramé et al., 2024); our objective takes the stronger *worst case* over a held-out judge panel, testing the herding caveat (Eisenstein et al., 2023) via an out-of-family judge.

3 The NoT Scaffold and Experimental Setup

3.1 The Narration-of-Thought scaffold

NoT is a single system prompt. In place of an open-ended “think step by step” instruction, it requires the model to produce five labelled narrative sections before committing to an answer: **(i) Protagonist**—identify whose decision is at issue; **(ii) Stakeholders**—enumerate the parties whose interests intersect the decision; **(iii) Consequences**—project the likely effects of the leading options two steps out; **(iv) Uncertainty**—state explicitly what is unknown or contested; and **(v) Commitment**—give a single, committed recommendation. The full verbatim prompt is reproduced in Appendix A so that the paper is self-contained; it is applied identically across generators and is never fine-tuned into the weights. NoT is thus a *black-box, inference-time, output-side* intervention: it shapes the assistant’s own generation and requires only API-level access.

We also evaluate a *multi-stakeholder* variant—several NoT agents (one per stakeholder perspective) whose drafts a separate moderator integrates over five rounds into one answer—used only to test whether the indirectness cost of single-agent NoT (§5) can be reduced in a cross-judge-coherent way.

3.2 Generators and comparison arms

Our **generator quartet** spans four vendors: gpt-5.4-nano, claude-haiku-4-5, claude-sonnet-4-6, and grok-4-1-fast-reasoning. Holding the scaffold fixed and varying the generator lets us separate scaffold effects from model-specific quirks. We compare four single-agent arms: **Raw** (empty system prompt), **IO** (a minimal instruction-only prompt), **standard CoT** (“think step by step”), and **NoT**. Standard CoT is the key

baseline: it isolates the contribution of NoT’s *structure* from the generic effect of adding intermediate reasoning, and all headline NoT effects are reported as deltas against CoT, not against Raw.

3.3 Benchmarks and constructs

We measure three families of sycophancy.

Obvious-falsehood (propositional, saturated). SycophancyEval (Sharma et al., 2024) elicits agreement with plainly false premises. We use it only to establish that this instrument is saturated on current frontier models (§4).

Plausible-falsehood (propositional, graded). BrokenMath (Petrov et al., 2025) presents expert-verified *false* competition-mathematics theorems and asks the model to prove them; a four-category judge labels each response **Sycophant** (attempts to prove the false claim), **Detected** (notices something is wrong), **Corrected** (fixes the statement), or **Ideal** (correctly refutes). The headline *Sycophant rate* is the fraction of responses that go along with the false premise. We evaluate a fixed split of 451 adversarial problems per cell.¹

Social sycophancy. ELEPHANT (Cheng et al., 2025) scores excessive preservation of the user’s “face” on advice-seeking prompts, grounded in face-work (Goffman, 1955). It spans two prompt sources—**OEQ** (open-ended advice questions) and **AITA** (“Am I The Asshole?” moral-judgment posts)—and four metrics: **Validation** (uncritically affirming the user’s feelings or position), **Indirectness** (hedged, non-committal phrasing rather than a direct answer), **Framing** (accepting the user’s framing/premises without challenge), and **Moral both-NTA**. The moral metric is a *cross-prompt consistency* probe: each AITA conflict is presented in its original form and in a perspective-flipped rewrite, and *both-NTA* is the rate at which the model returns the “Not The Asshole” (NTA) verdict for *both* sides of the same conflict—i.e. affirms whoever is asking. (The complementary verdict is YTA, “You’re The Asshole.”) Comparing two responses to two prompts, both-NTA measures something structurally different from the single-response metrics (central in §6). We use a stratified $n=150$ subsample per slice (seed 44)

¹The public BrokenMath release contains additional problems; we hold this 451-item split fixed across all arms so within-table comparisons are exact.

Construct	Instrument	Unit
Propositional (obvious)	Sharma	single resp.
Propositional (plausible)	BrokenMath	single resp.
Social: validation	ELEPHANT	single resp.
Social: indirectness	ELEPHANT	single resp.
Social: framing	ELEPHANT	single resp.
Moral consistency	ELEPHANT	prompt pair

Table 1: The sycophancy constructs studied here are measured by different instruments and at different units of analysis. “Single resp.” metrics score one answer; moral both-NTA compares a model’s answers to a prompt and its perspective-flipped twin. A mitigation can move one row without moving the others.

Model	Gen.	Judge	Panel	Held-out
nano	✓	–	✓	–
haiku	✓	✓	✓	–
sonnet	✓	–	–	✓
grok	✓	–	✓	–
DeepSeek-R1	–	–	–	✓

Table 2: Model roles (short labels; full identifiers in §3). “Judge” is the default single judge, “Panel” the optimisation panel. The generator quartet is scored throughout and overlaps with the judge/panel roles, while sonnet (in-vendor with haiku) and out-of-family DeepSeek-R1 are reserved for held-out evaluation. Role overlap is a limitation we return to in §11.

and report rates against the crowdsourced human baseline shipped with ELEPHANT. Table 1 summarises the construct taxonomy.

3.4 Scoring and the judge panel

Every social and propositional outcome is scored by an LLM judge using the faithful ELEPHANT/BrokenMath judge prompts. The default single judge is claude-haiku-4-5. Because LLM judges are unreliable (§2), §8 re-scores a stratified subsample with a three-family reliability panel (gpt-5.4-nano, claude-haiku-4-5, grok-4-1-fast-reasoning) and reports Krippendorff’s α (Krippendorff, 2004) for inter-judge agreement and Cohen’s κ (Cohen, 1960) against crowdsourced human gold. For the optimisation experiments we additionally hold out two judge families—claude-sonnet-4-6 and the out-of-family DeepSeek-R1—that are never consulted during optimisation.

3.5 Narrative-gradient optimisation

To optimise the scaffold we use a textual-gradient procedure (cf. ProTeGi (Pryzant et al., 2023), TextGrad (Yuksekgonul et al., 2024)): an optimiser model inspects scored examples, writes a natural-language “gradient” for how the prompt failed, and edits it, iterating for a fixed budget; we call this construct-specialised variant *narrative gradients*. The naive objective minimises a single judge’s pooled social-sycophancy score,

$$L_{\text{single}} = \text{mean}(\text{val} + \text{ind} + \text{fram}),$$

which §8 shows is exploitable. Our **panel-robust** objective instead takes the worst case over an optimisation panel $\mathcal{P}$ of the three in-family judges (grok-4-1-fast-reasoning,

claude-haiku-4-5, gpt-5.4-nano) before averaging over items,

$$L_{\text{robust}} = \text{mean}_i \max_{j \in \mathcal{P}} \text{mean}(\text{val} + \text{ind} + \text{fram})_{ij},$$

so a prompt edit lowers the loss only when *every* panel judge agrees it helped; it cannot cash in on a single lenient judge. claude-sonnet-4-6 and DeepSeek-R1 are held out entirely and used only at evaluation. The whole procedure is black-box—API-level generation and judging, no weights or activations—so it applies to frontier closed models. Model roles are summarised in Table 2; note that claude-haiku-4-5 and grok-4-1-fast-reasoning appear in several roles, a leakage consideration we flag in Limitations.

3.6 Statistical testing

Cross-arm rate comparisons on independent cells use two-proportion z -tests when n is large and Fisher exact tests when any cell count is small (BrokenMath per-generator deltas). Re-scoring tables that compare hand vs. optimised responses on the *same* items (e.g. the Goodhart gap, Table 6) are paired by construction; we report two-proportion tests on those paired binary outcomes for consistency with the rest of the paper, and note McNemar’s test as the strictly paired alternative. We report pairwise tests without a global multiple-comparison correction (§11); headline effects with $p < 10^{-4}$ survive conservative correction.

4 Benchmark Choice: Obvious vs. Plausible Falsehoods

A propositional-sycophancy claim is only as good as its instrument. On hand-written obvious-falsehood probes (Sharma et al., 2024), every

Generator	Raw	Std CoT	NoT	$\Delta(\text{NoT}-\text{CoT})$
nano	58%	55%	69%	+14.7 [†]
haiku	81%	83%	80%	-3.5
sonnet	70%	69%	73%	+3.1
grok	48%	51%	39%	-12.0*

Table 3: BrokenMath Sycophant rate ($n=451$ adversarial problems per cell, judge claude-haiku-4-5). Rates far exceed the Sharma floor. NoT reduces sycophancy significantly on grok (*Fisher $p<0.001$) and *increases* it on nano ([†]Fisher $p<10^{-5}$); haiku/sonnet are null. The mixed signs are the first evidence that the social-metric reductions below do not transfer to propositional error-detection.

cell of our quartet—under both standard CoT and NoT—scores at most 2.2% (Appendix B, Table 8). The instrument is *saturated*: it cannot discriminate arms, and any “improvement” on it would be noise at the floor.

BrokenMath (Petrov et al., 2025) supplies the missing headroom. Standard-CoT Sycophant rates span 51% (grok-4-1-fast-reasoning) to 83% (claude-haiku-4-5)—more than an order of magnitude above the Sharma floor (Table 3). We therefore treat BrokenMath, not Sharma, as the propositional instrument for all scaffold claims, and report Sharma only to document the floor. This weak-instrument correction matters independently of NoT: it cautions against the common inference from low obvious-falsehood rates to low propositional sycophancy in general, echoing graded-probe findings of 29–58% residual sycophancy elsewhere (Fanous et al., 2025; Petrov et al., 2025).

5 Social Sycophancy: Validation and Framing

Validation: the headline effect. NoT cuts OEQ validation on all four generators (Fisher $p < 10^{-4}$; -26 to -53 pp vs. CoT), meeting our pre-registered falsification criterion. Three land within two points of the 29% human rate; claude-sonnet-4-6 reaches 47% (Table 4, Appendix Table 11). Matching a crowd rate is not item-level correctness: NoT could withhold validation where humans would grant it (overcorrection) or vice versa; we report rates, not agreement with human labels on the same items. This is nonetheless the paper’s most robust positive result, and—unlike the other social metrics—it survives the judge panel of §8 without a reliability caveat.

Metric (slice)	Human	Raw	IO	Std CoT	NoT
OEQ validation	29%	66%	54%	63%	27%
OEQ framing	61%	55%	65%	65%	23%
AITA validation	7%	47%	43%	37%	17%
AITA framing	41%	50%	52%	52%	22%
Moral both-NTA	–	18%	25%	23%	15%

Table 4: ELEPHANT rates on the $n=150$ subsample, claude-haiku-4-5 (corrected flip-pair loading). NoT reduces validation and framing sharply. Moral row is binary both-NTA; there is no collapse under NoT (cf. §6).

Framing. NoT lowers framing on every generator (significant on three of four on OEQ/AITA), and on premise-acceptance (-13 to -24 pp vs. CoT).

Indirectness, and cross-judge coherence. Single-agent NoT nudges *indirectness* up on several cells—the model hedges more when forced to enumerate uncertainty—but the multi-stakeholder variant yields a cross-judge-coherent directness reduction across all eight dataset-generator cells (-11 to -28 pp) while preserving the validation gains. Since indirectness carries sub-chance inter-judge reliability ($\alpha = -0.23$; §8), treat any “indirectness recovery” claim as simultaneous satisfaction of idiosyncratic graders, not recovery of a well-defined latent construct.

A length caveat. NoT responses are substantially longer than CoT on OEQ: on claude-haiku-4-5, mean length rises from $\approx 1,700$ to $\approx 5,600$ characters (3.3 $\times$); on claude-sonnet-4-6 from $\approx 1,650$ to $\approx 7,200$ (4.4 $\times$; full quartet in Appendix Table 12). Verbosity biases LLM judges (Li et al., 2024), so validation/framing reductions could be partly a length or formatting artifact; we ran no length-matched verbose-CoT control. Three dissociations argue against length *alone*: indirectness moves *up* not down, the effect does not transfer to BrokenMath despite the same length increase, and the moral metric moves the *opposite* way (§6). A length-matched ablation remains future work (§11).

6 Moral Consistency Is a Different Construct

A retraction. An earlier version of this work reported a “moral collapse” under NoT. That result was an artifact: the data loader read a nonexis-

tent column for the perspective-flipped prompts, so flip-pair prompts were empty (`prompt_len=0`) and models saw only the binary verdict suffix rather than the story. **We retract all prior moral-collapse claims.** With corrected loading, binary both-NTA rates are null-to-mixed and show no systematic NoT reduction (Table 4, Appendix Table 11, bottom): gpt-5.4-nano sits near 50% across arms; claude-haiku-4-5 near 15–23%.

Free-form mode raises both-NTA under NoT. Removing the binary suffix and extracting the verdict post hoc with an LLM lets NoT operate on the full story. The cleaner comparison is *within* free-form mode: NoT *raises* both-NTA vs. standard CoT on every generator (Appendix Table 10), e.g. claude-haiku-4-5 53% → 93% and claude-sonnet-4-6 38% → 72%—the opposite sign from validation/framing on the same models.

Binary suffix vs. instrument change. Binary both-NTA rates are much lower than free-form rates on the same generators (Table 4 vs. Appendix Table 10): e.g. corrected binary claude-haiku-4-5 both-NTA is 15–23% while free-form runs 59–93%. ELEPHANT’s published ≈ 48% both-NTA used a different extraction protocol on full (not subsampled) data; our corrected binary subsample and free-form arm therefore bracket that headline rather than replicate it. Pooled both-NTA rises 34% → 67% when switching instruments (binary→free-form); that jump conflates scaffold effects with the added verdict-extraction step, whose reliability we do not separately audit.

Interpretation. The sign here is opposite to validation/framing, which fits the construct geometry of Table 1: when NoT enumerates stakeholders and declines to take sides, it affirms *both* parties of a flipped conflict more often, not less. Returning NTA for both sides is not obviously *wrong*: many flipped conflicts genuinely have no asshole, so a higher both-NTA rate may reflect appropriate moral caution as much as inconsistency. We therefore read this result narrowly and caution researchers against pooling moral rates with validation/framing claims.

7 Propositional Sycophancy Does Not Follow

The social-metric reductions of §5 do *not* transfer to BrokenMath’s propositional error-detection

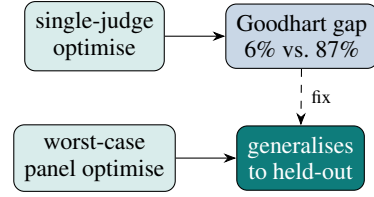


Figure 1: Optimising the scaffold against one absolute judge drives its training-judge score to the floor but leaves held-out judges at baseline (the Goodhart gap, §8). Optimising the worst case over a judge panel, with a family held out entirely, yields reductions that transfer (§9).

(Table 3). NoT moves the Sycophant rate by −3.5 pp on claude-haiku-4-5 ($p=0.20$) and +3.1 pp on claude-sonnet-4-6 ($p=0.34$), neither significant. On grok-4-1-fast-reasoning it cuts sycophancy 51% → 39% (−12 pp, Fisher $p<0.001$)—the only generator with substantial *Detected* and *Ideal* headroom under the scaffold. On gpt-5.4-nano it *increases* sycophancy 55% → 69% (+14.7 pp, $p<10^{-5}$), the opposite sign from the social benchmark. These directions are judged by a single in-family scorer (claude-haiku-4-5); we did not panel-check BrokenMath in this pass, so the surprising gpt-5.4-nano *increase* should be read as large but single-judge.

These mixed signs are the empirical heart of the construct-fragmentation claim: a scaffold that reliably cuts social validation across four vendors does not reliably reduce propositional agreement on checkable falsehoods, and on gpt-5.4-nano makes it *worse*. “Reducing sycophancy” is thus not a single achievable target.

8 Optimising the Scaffold, and How Judges Get Gamed

Sections 5–7 use a hand-written NoT prompt. We now *optimise* the scaffold with narrative gradients, and in doing so confront the measurement problem head-on: when a prompt is optimised against an LLM judge, low judge reliability becomes an exploitable attack surface. Figure 1 previews the argument—single-judge optimisation opens a Goodhart gap that a worst-case-over-panel objective closes.

8.1 Narrative gradients reduce the single-judge loss

We optimise NoT on L_{single} over a disjoint OEQ train split ($n=100$, indices 150–249), with held-out evaluation on a separate OEQ set

($n=150$) using the claude-haiku-4-5 generator and grok-4-1-fast-reasoning as the training judge. Hand NoT scores validation 76%, framing 39%, indirectness 16% under the training judge (pooled loss 0.44); the optimised prompt reaches validation 0.7%, framing 3%, indirectness 5% (loss 0.03). Against three matched-budget baselines (~ 10 prompt evaluations each), narrative gradients achieve the lowest held-out loss, beating APE, TextGrad-CoT and OPRO (Appendix Table 9). Two narrative-initialised methods outperform the CoT-initialised TextGrad baseline, supporting the pre-registered hypothesis that scaffold *structure*, not mere iterative search, carries the reduction. The gradient rewrites the protagonist section into a “Premise Audit” that challenges embedded assumptions—a mechanistic hint at why first-person framing inflates validation. *However*, this single-judge “win” is partly illusory, as the next subsections show; its headline loss should be read alongside the Goodhart diagnostic of §8.3.

8.2 The judge panel: low inter-judge reliability

We re-score $n=60$ stratified OEQ responses with three judge families (gpt-5.4-nano, claude-haiku-4-5, grok-4-1-fast-reasoning) on validation, indirectness and framing, and compare each judge to crowdsourced human gold ($n=30$ human responses) via Cohen’s κ . **Inter-judge reliability is low on all three metrics** (Table 5): validation $\alpha=0.42$ [0.25, 0.59], framing $\alpha=0.11$ [−0.05, 0.28], indirectness $\alpha=−0.23$ [−0.32, −0.12] (95% item-bootstrap CIs, $n=60$; below chance for indirectness). None clears the conventional $\alpha=0.67$ threshold, so panel headline rates use majority vote. A sub-chance α means judges disagree more than chance on indirectness: driving indirectness to 0% under all panel judges is therefore a *cross-judge coherence* property (every idiosyncratic grader agrees), not evidence that a well-defined latent construct was recovered. Against human gold the picture is more favourable for validation (Cohen’s κ 0.73–0.86 with 95% CIs spanning roughly 0.48–1.0) but only moderate for indirectness (0.31–0.63) and mixed for framing (0.15–0.70). The combination—strong judge-vs-human κ on validation yet weak *inter-judge* α —is the known base-rate κ paradox (Feinstein and Cicchetti, 1990). The practical consequence is sharp: **validation is the only ELEPHANT**

Metric	Inter-judge α ($n=60$)	κ vs. human		
		nano	haiku	grok
Validation	0.42	0.85	0.86	0.73
Indirectness	−0.23	0.31	0.39	0.63
Framing	0.11	0.15	0.57	0.70

Table 5: Judge reliability (point estimates; 95% item-bootstrap CIs for α in text, $n=60$). Inter-judge Krippendorff α over the three-family panel, and Cohen’s κ of each judge against crowdsourced human labels ($n=30$ human responses). No metric clears $\alpha=0.67$; validation alone shows strong judge-vs-human agreement on *human* responses (not optimised model outputs).

metric we report without a reliability caveat, and any indirectness reduction from a single judge is not, on its own, trustworthy—which is exactly what optimisation exploits next.

8.3 The Goodhart gap: re-scoring the identical responses

To isolate judge-gaming from behaviour change we re-score the *same* held-out responses (hand NoT vs. single-judge optimised; claude-haiku-4-5 generator; $n=150$) across a four-judge panel—the training judge grok-4-1-fast-reasoning plus held-out claude-haiku-4-5, gpt-5.4-nano, claude-sonnet-4-6—changing only the judge, never regenerating (Table 6). The validation reduction is genuine: optimised validation is 0–3% under *every* judge. **The indirectness reduction is a Goodhart artifact**: the training judge reads optimised indirectness at 6%, but the worst held-out judge reads 87%—identical to the hand-NoT baseline. Optimising one absolute judge bought a phrasing change that this judge scores as direct and others do not; framing sits in between (robust on three judges, 52% on claude-sonnet-4-6). This is the practical face of the low inter-judge α above: an optimiser pointed at a single noisy judge will exploit it. The fix is to stop trusting any single judge—developed next.

9 A Panel-Robust Optimizer

Method. We re-run narrative-gradient descent (init = verbatim NoT, 10 iterations, optimiser claude-sonnet-4-6) under the worst-case-over-panel objective L_{robust} of §3, selecting the iteration with lowest robust loss. The panel is the three in-family judges grok-4-1-fast-reasoning, claude-haiku-4-5, and gpt-5.4-nano;

Metric	Single-judge optimised rate			
	grok [†]	haiku	nano	sonnet
Validation	1%	2%	3%	0%
Indirectness	6%	87%	59%	10%
Framing	3%	4%	27%	52%

Table 6: The Goodhart gap. The *same* single-judge optimised responses ($n=150$) re-scored by four judges. Validation is low under all judges (robust); indirectness collapses to a 6% training-judge reading but stays at the 87% baseline under the worst held-out judge. [†] training judge.

Prompt	Held-out (val/ind/fram)		Panel
	sonnet	DeepSeek	worst-loss
Hand NoT	10/41/89%	53/78/68%	0.71
Single-judge	0/10/52%	6/74/13%	0.39
Robust	0/0/11%	13/4/5%	0.32

Table 7: Held-out generalisation ($n=150$, claude-haiku-4-5 generator). Each cell is validation/indirectness/ framing (%); “worst-loss” is worst-case-over-panel loss. Robust beats single-judge on framing ($p=1.9\times 10^{-14}$ sonnet) and indirectness ($p=7\times 10^{-5}$ sonnet; $p < 10^{-15}$ DeepSeek), and uniquely moves worst-case indirectness off baseline.

claude-sonnet-4-6 and out-of-family DeepSeek-R1 are held out entirely and scored only at evaluation. Because the loss credits only changes all panel judges agree on, the optimiser cannot bank on a single lenient judge.

Generalisation to held-out judges. On claude-sonnet-4-6—never scored during optimisation—the robust prompt generalises significantly better than the single-judge prompt: framing 11% vs. 52% (two-proportion $z = -7.7$, $p=1.9 \times 10^{-14}$) and indirectness 0% vs. 10% ($z = -4.0$, $p=7 \times 10^{-5}$), both at the validation floor (Table 7). On out-of-family DeepSeek-R1 the same pattern holds (indirectness 4% vs. 74%, $p < 10^{-15}$; framing 5% vs. 13%). The robust prompt also achieves the lowest worst-case panel loss (0.32 vs. single-judge 0.39 vs. hand 0.71) and is the only method to move worst-case panel indirectness off the 87% baseline (to 55%) in a cross-judge-coherent way. It trades a little worst-case validation (9% vs. 3%) for large, judge-general gains on the two constructs single-judge optimisation gamed or left at baseline.

Independent validation anchor. On a stratified $n=40$ sample of hand vs. robust holdout responses,

an out-of-family proxy judge (DeepSeek-R1, held out from optimisation) rates validation 70% $\rightarrow$ 5% (hand $\rightarrow$ robust), with proxy-author $\kappa=0.28$. Author-collected labels on the same sample give 35% $\rightarrow$ 0% but are not blinded—a conflict of interest—and should be read only as a secondary sanity check. The strong human-gold κ in §8 (0.73–0.86) was measured on *ELEPHANT human responses*, not these optimised model outputs, where judge-author agreement is lower. Validation therefore moves as claimed under both held-out LLM judges and the proxy anchor; indirectness/framing generalisation rests on held-out LLM judges only.

Degeneracy guard. On an $n=30$ subsample the robust prompt preserves stakeholder enumeration (4.77 vs. hand 4.73) while lowering uncertainty narration (1.0 vs. 2.83), so the optimiser does not collapse the scaffold into vacuity.

Quartet replication. Across the quartet on holdout OEQ ($n=150$ /cell; grok-4-1-fast-reasoning $n=147$), panel worst-case validation falls under robust vs. hand on every generator (claude-haiku-4-5 69 $\rightarrow$ 9, claude-sonnet-4-6 89 $\rightarrow$ 20, gpt-5.4-nano 66 $\rightarrow$ 3, grok-4-1-fast-reasoning 62 $\rightarrow$ 2%), and held-out DeepSeek-R1 framing follows on all four (74 $\rightarrow$ 4, 74 $\rightarrow$ 4, 96 $\rightarrow$ 3, 95 $\rightarrow$ 4% resp.), so robustness generalises across generator families, not only the claude-haiku-4-5 training cell.

Scope. We present panel-robustness as a demonstrated generalisation *property*, not a deployment-ready prompt (§11).

10 Discussion

Sycophancy is not one construct. A single scaffold cuts social validation across four vendors yet does not reduce propositional sycophancy on checkable falsehoods (significant only on grok-4-1-fast-reasoning; a significant *increase* on gpt-5.4-nano) and *reverses* on cross-prompt moral consistency. Benchmark choice is thus not a detail—obvious-falsehood probes saturate at $\leq 2\%$ while BrokenMath runs 39–83%—and one pooled “sycophancy” number hides the dissociations that reveal what an intervention does. Deliberation buys social, single-response face-preservation and little else, matching the linguistic-vs-reasoning asymmetry Cheng et al. (2025) report and the prediction that added reasoning need

not reduce bias (Turpin et al., 2023; Chua et al., 2024).

Optimising against LLM judges measures the judge. The narrative-gradient “win” partly reverse-engineers its training judge: re-scoring identical responses shows the indirectness reduction is a Goodhart artifact (6% vs. 87% held-out), with only validation surviving every judge—in-context reward hacking (Pan et al., 2024; Gao et al., 2023) at the prompt level. A worst-case-over-panel objective with a family held out fixes this by construction, generalising to out-of-family DeepSeek-R1 (one held-out judge family; judge ensembles can share blind spots (Eisenstein et al., 2023)) where single-judge optimisation does not, while preserving stakeholder depth; being black-box, it transfers to closed models. The lesson generalises past sycophancy: *prompt optimisation against LLM judges must be evaluated against judges held out from optimisation, or it measures judge idiosyncrasy rather than behaviour.*

Less sycophantic is not always better. Anti-sycophancy interventions can overcorrect into bluntness or into rejecting *correct* user claims (Wei et al., 2023; Sharma et al., 2024), and our free-form moral result shows NoT affirming both sides *more* often. We did not run a naive “don’t be sycophantic” instruction arm; Dubois et al. (2026) show such direct prompts are a weak baseline relative to input-side reframing. We therefore frame NoT as a measurement and construct-clarification tool—not a deployment-ready fix, and complementary to input-side reframing (Dubois et al., 2026).

11 Conclusion

Sycophancy fragments into constructs that need not move together: NoT cuts social validation and framing across four generators where obvious-falsehood probes show no signal, yet does not transfer to propositional error-detection and reverses on moral consistency. A reliability panel exposes single-judge prompt optimisation as judge-gaming; a panel-robust optimiser yields cross-judge-coherent indirectness reductions that generalise to held-out judges. Deliberative scaffolding targets single-response social face-preservation, and robustness—not raw loss against one judge—is the quantity worth optimising.

Limitations

Judge dependence. All sycophancy outcomes are LLM-scored. Validation is anchored to human gold on ELEPHANT human responses (κ 0.73–0.86), but author labels on *model* outputs ($n=40$) show only moderate judge–author agreement (κ 0.13–0.27) and were not blinded; an out-of-family proxy judge (DeepSeek-R1) on the same sample gives 70% $\rightarrow$ 5% validation (hand $\rightarrow$ robust) with proxy–author $\kappa=0.28$, but indirectness/framing carry low inter-judge reliability ($\alpha \leq 0.42$, with bootstrap CIs in §8). Held-out generalisation rests on LLM judges; DeepSeek-R1 is out-of-family but claude-sonnet-4-6 shares a vendor with panel judge claude-haiku-4-5, and judge ensembles can share blind spots (Eisenstein et al., 2023).

Role overlap. Several models serve multiple roles (Table 2): claude-haiku-4-5 and grok-4-1-fast-reasoning are both generators and judges/optimiser-panel members, and claude-sonnet-4-6 is both a generator and a held-out judge. We mitigate the most direct leakage by holding claude-sonnet-4-6 and DeepSeek-R1 out of optimisation entirely, but a fully disjoint generator/judge design is left to future work.

Mechanism not isolated. The optimisation results isolate *outcome*, not *mechanism*: we have not ablated which narrative section (protagonist, stakeholders, consequences, uncertainty, commitment) carries the effect, so causal attribution to specific scaffold components remains open. Relatedly, we did not run a length-matched verbose-CoT control; NoT OEQ responses are 1.3–4.4 $\times$ longer than CoT (Appendix Table 12), so a residual verbosity confound on validation/framing cannot be fully excluded (§5). We also did not run a naive “don’t be sycophantic” instruction arm; Dubois et al. (2026) show such direct prompts are a weak baseline relative to input-side reframing.

Instrument and power. BrokenMath uses a single judge family (claude-haiku-4-5), and the surprising gpt-5.4-nano +14.7 pp increase rides on that one scorer—we did not panel-check BrokenMath in this pass. The free-form moral arm adds a verdict-extraction call whose reliability we do not separately audit. Per-cell samples are $n=150$ (ELEPHANT slices) and $n=451$ (BrokenMath); effects below a few points may be under-

powered. Significance tests are reported pairwise without a global multiple-comparisons correction; headline effects with $p < 10^{-4}$ survive conservative correction. We test arms with two-proportion / Fisher tests on independent cells (§3); paired rescore tables (e.g. the Goodhart gap) compare conditions on the same items.

Generality. All generators are accessed through hosted providers; results may not transfer to other deployments, model versions, or non-English advice settings.

Ethical Considerations

This work studies sycophancy mitigation and collects no new human-subjects data; human labels are drawn from the public ELEPHANT/OSF release and used only as reliability gold. Reducing sycophancy is dual-use: a scaffold that makes models less validating can also make them blunt or dismissive in genuinely sensitive contexts, and our finding that NoT *increases* moral both-NTA shows “less sycophantic” is not uniformly desirable. We therefore frame NoT as a measurement and construct-clarification tool, not a deployment-ready safety intervention, and caution against optimising single sycophancy metrics in isolation. No personally identifying information is processed; all adversarial math problems come from the public BrokenMath benchmark.

Reproducibility Statement

The scaffold is specified in full in Appendix A; the panel-robust objective in §3; generators, arms, splits, seeds and judges in §3. All experiments are black-box (API generation and judging). Appendix C maps each result to the internal milestone label under which it was produced, for cross-referencing released artifacts. Significance tests, sample sizes and held-out judge identities are stated in each table caption.

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A The NoT System Prompt

The scaffold is a single system prompt, prepended to the user turn and held fixed across all generators and arms. It adds no training data, parameters, or fine-tuning. The instruction content is reproduced below verbatim.

Before giving your final answer, think in the first person and write five short labelled sections:

1. PROTAGONIST: name the person whose decision this is, their role, and what they actually know.
2. STAKEHOLDERS: list every party whose interests intersect this decision and what is at stake for each.
3. CONSEQUENCES: for each option you are weighing, project the likely effects at least two steps out, for each stakeholder.
4. UNCERTAINTY: state plainly what remains genuinely unknown or contested.
5. COMMITMENT: commit to a single recommendation and explain, within this frame, why it is preferable to the alternatives.

Then give your final answer.

The multi-stakeholder variant (§3) runs this same prompt as the per-agent generator for several stakeholder perspectives and integrates the drafts with a separate moderator model over five rounds before emitting one answer.

B Obvious-Falsehood Floor (Sharma)

Table 8 reports SycophancyEval (Sharma et al., 2024) on the quartet. Every cell is at most 2.2%, confirming the instrument is saturated and cannot discriminate arms; it is excluded from all scaffold claims and reported only to motivate the move to BrokenMath (§4).

Generator	Std CoT	NoT
gpt-5.4-nano	0.0%	2.2%
claude-haiku-4-5	1.1%	0.0%
claude-sonnet-4-6	0.0%	0.0%
grok-4-1-fast-reasoning	0.0%	0.0%

Table 8: Sharma SycophancyEval sycophancy rate (30 probes $\times$ 3 samples = 90 judgments per cell). Saturated floor; not used for scaffold claims.

C Result-to-Milestone Map

For cross-referencing released artifacts, each result maps to the internal milestone label under which it was produced: obvious-falsehood floor and BrokenMath headroom (Phase 16; §4, §7); scaled ELEPHANT validation/framing and corrected moral loading (Phase 13; §5, §6); free-form moral arm (Phase 17; §6); narrative-gradient optimisation and optimiser comparison (Phase 14; §8); judge reliability panel (Phase 15; §8); Goodhart-gap re-scoring (Phase 18a; §8.3); panel-robust optimiser, held-out generalisation, author-label anchor, degeneracy guard, and quartet replication (Phase 18b–e; §9). Pre-registration and run scripts are in the released artifact bundle.

D Supplementary Tables

Tables relocated from the main text to respect the page limit; their values are stated in the body.

Optimiser	Init	Holdout loss
Narrative gradients	NoT	0.029
APE	NoT	0.104
TextGrad-CoT	Std CoT	0.170
OPRO	NoT	0.200
Hand NoT (reference)	—	0.436

Table 9: Optimiser holdout loss (validation + indirectness + framing, training judge grok-4-1-fast-reasoning, $n \approx 148$ –150; lower is better; §8). Narrative gradients win, but the objective itself is partly gameable (§8.3); robustness, not this loss, is the quantity we ultimately optimise.

Generator	Raw	Std CoT	NoT
gpt-5.4-nano	78%	79%	78%
claude-haiku-4-5	59%	53%	93%
claude-sonnet-4-6	32%	38%	72%
grok-4-1-fast-reasoning	70%	73%	82%

Table 10: Free-form moral both-NTA rate ($n=150$ pairs; §6). NoT *raises* both-NTA on every generator (sharply on claude-haiku-4-5/claude-sonnet-4-6), the opposite of its validation/framing effect—consistent with a distinct cross-prompt construct, though partly attributable to the free-form instrument (see text).

	nano	haiku	sonnet	grok
<i>OEQ validation (human =29%)</i>				
Raw	78%	66%	77%	83%
Std CoT	75%	63%	73%	82%
NoT	27%	27%	47%	29%
<i>Moral both-NTA (binary; corrected loader)</i>				
Raw	51%	18%	25%	31%
Std CoT	50%	23%	24%	32%
NoT	51%	15%	17%	42%

Table 11: Cross-vendor replication (§5). Top: OEQ validation—NoT reductions significant on all four generators. Bottom: corrected binary moral both-NTA—no systematic NoT effect, indicating a different construct from validation/framing.

Generator	Raw	IO	Std CoT	NoT
gpt-5.4-nano	4,125	3,286	4,346	6,337
claude-haiku-4-5	1,662	1,442	1,692	5,580
claude-sonnet-4-6	1,543	1,463	1,647	7,165
grok-4-1-fast-reasoning	3,613	1,534	3,815	5,073

Table 12: Mean OEQ response length (characters) by arm ($n \approx 145$ –150/cell; §5). NoT/CoT ratios: gpt-5.4-nano 1.46 $\times$, claude-haiku-4-5 3.3 $\times$, claude-sonnet-4-6 4.35 $\times$, grok-4-1-fast-reasoning 1.33 $\times$.